

A 3D Mosque

Submitted To:

Dr. Sk. Md. Masudul Ahsan
Professor
Department of CSE, KUET

Md Tajmilur Rahman
Lecturer

Department of CSE, KUET

Submitted By:

Shoab Ahamed
Roll: 2007015
Year: 4th
Semester: 2nd

Department of Computer Science and Engineering
Khulna University of Engineering & Technology
Khulna -9203, Bangladesh.

Introduction

Basic Objects of the project are:

1. Cube
2. Sphere
3. Cylinder
4. Cone
5. Bezier Curve
 - Mirhab
 - Chandelier

Introduction

The components of the mosque

- | | |
|----------------|-----------|
| 1. Trees | 8. Roads |
| 2. Wadhu Place | 9. Mirhab |
| 3. Bookselves | |
| 4. Fans | |
| 5. Ponds | |
| 6. Dome | |
| 7. Minarets | |

Functional Description

1. General Controls

- ESC → Close the window

2. Camera Movement

- W → Move forward
- S → Move backward
- A → Move left
- D → Move right
- R → Move down
- E → Move up

3. Camera Rotation (Yaw, Pitch, Roll)

- Y → Rotate Y axis (positive)
- Shift + Y → Rotate Y axis (negative)
- X → Rotate X axis (look up)
- Shift + X → Rotate X axis (look down)
- Z → Rotate Z axis (positive)
- Shift + Z → Rotate Z axis (negative)

Functional Description (contd.)

5. Dynamic Elements

- B → Bookself open/close
- H → Toggle faucet **water droplets**
- O → Door open/close
- P → Window open/close
- 4 → Fan on/off

6. Lighting Controls

- 1 → Toggle Directional Light
- 2 → Toggle Point Light
- 3 → Toggle Spot Light
- 5 → Toggle Ambient lights on/off
- 6 → Toggle Diffuse light on/off
- 7 → Toggle Specular light on/off

Detailed Visualization

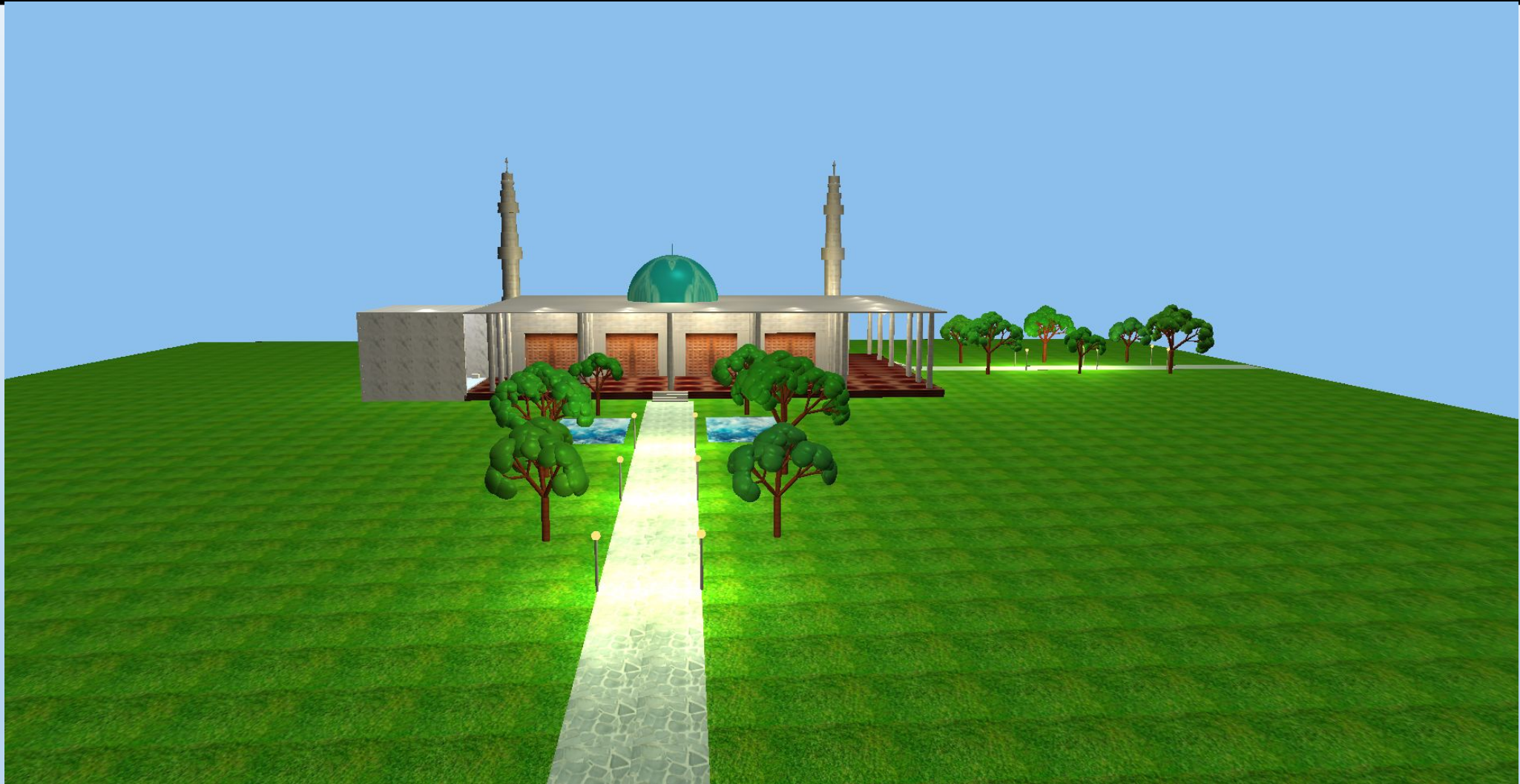


Fig-1: Overall View of the project: All lights on

Detailed Visualization

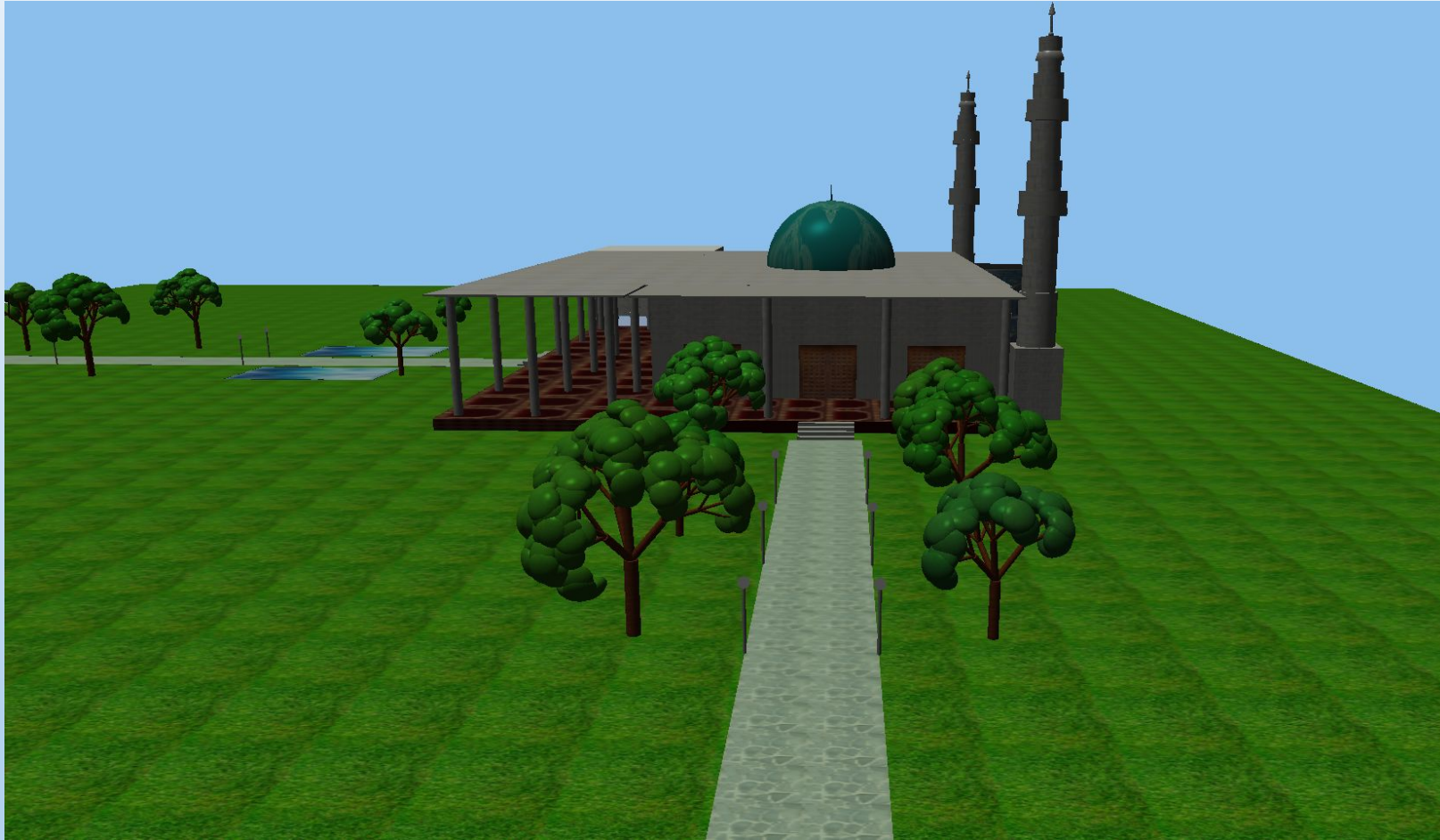


Fig-2: Overall View of Project: Point lights off

Detailed Visualization



Fig-1: Overall View of Project: Directional light off

Detailed Visualization

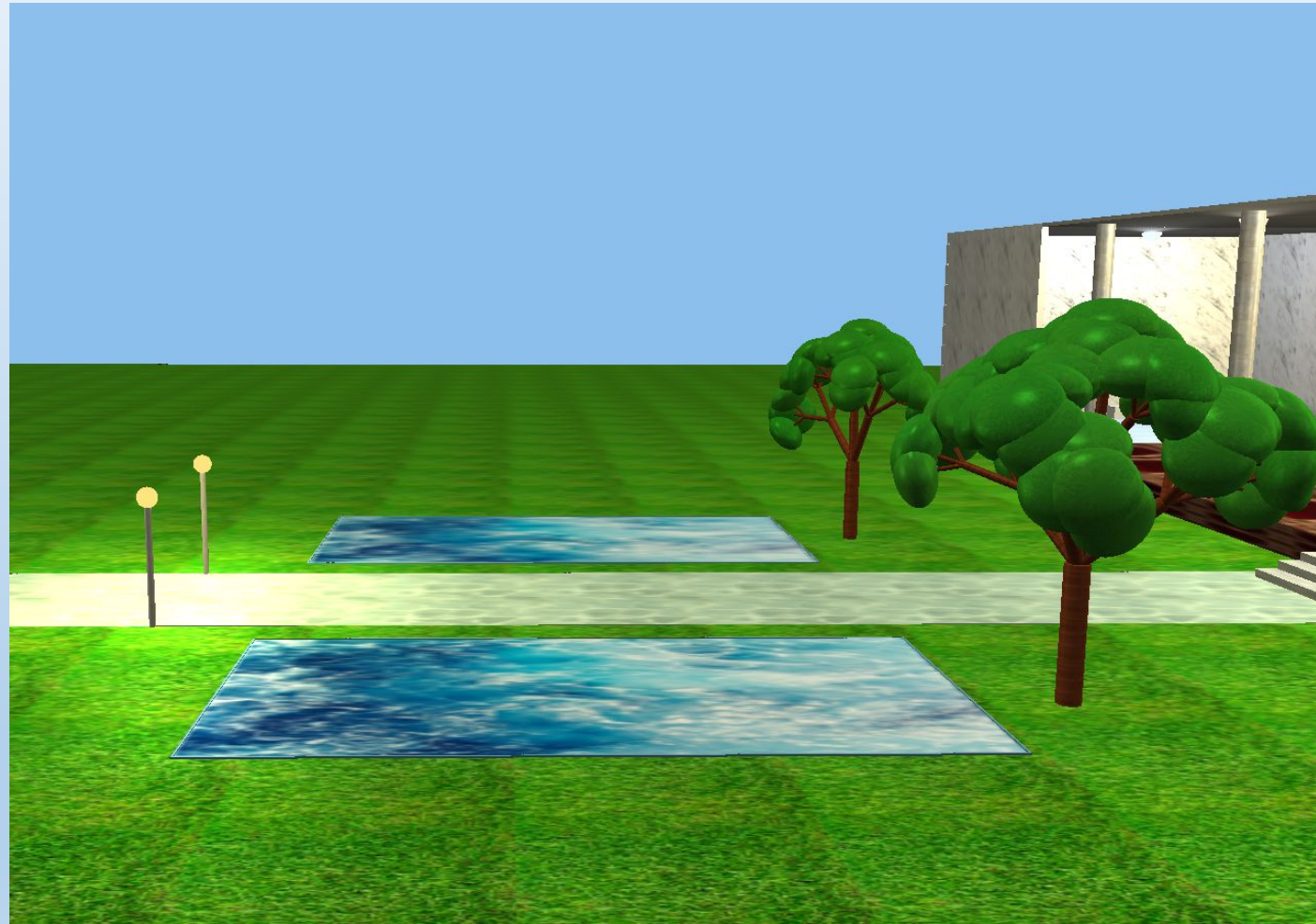


Fig-4: Ponds & Trees

Detailed Visualization

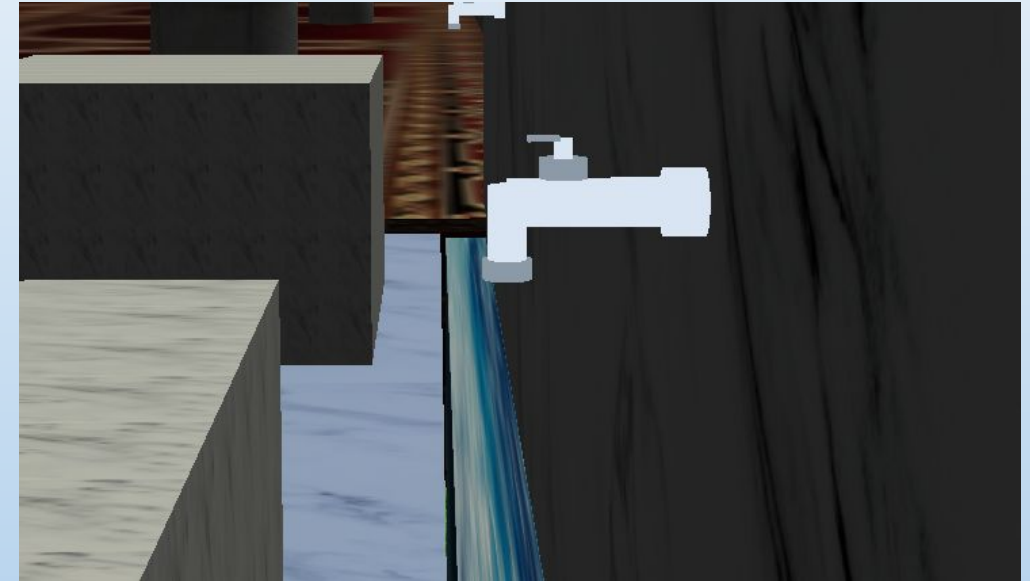


Fig-5: Wadhu place & Faucet

Detailed Visualization



Fig-6: Faucets & Droplet Animation

Detailed Visualization



Fig-7: Doors open and closed

Detailed Visualization

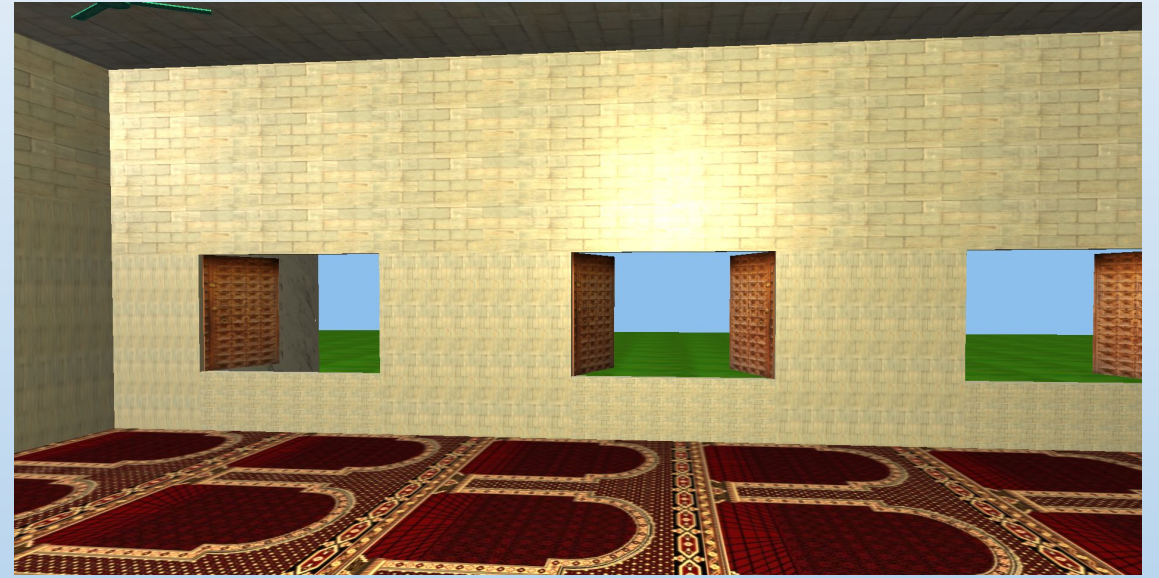


Fig-8: Windows open & Closed

Detailed Visualization



Fig-9: Mosque Interior

Detailed Visualization



Fig-10: Mosque Interior: Spotlight On

Detailed Visualization



Fig-11: Alcove Bookself

Detailed Visualization

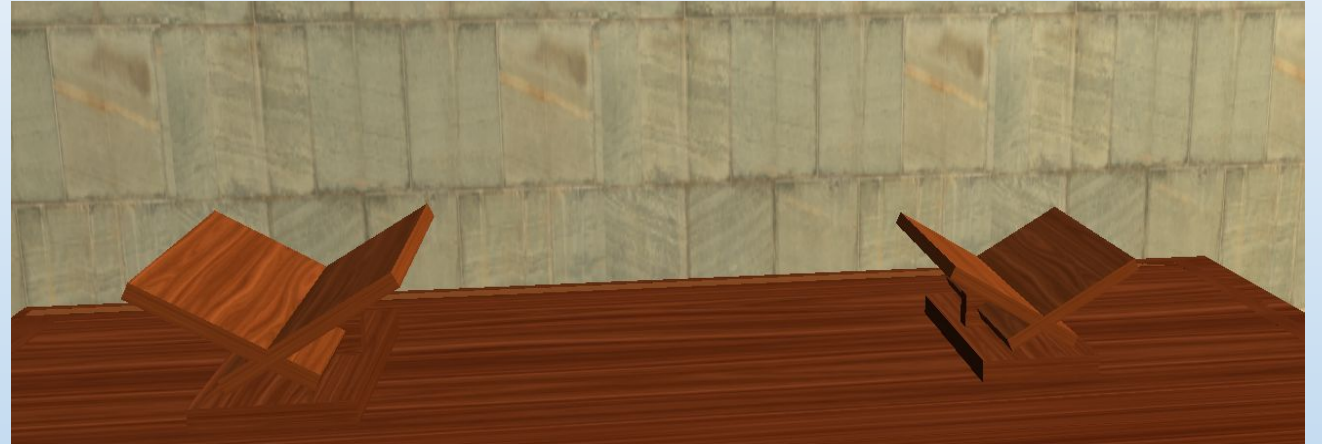


Fig-12: Openable Regular Bookself & Rehal

Detailed Visualization



Fig-13: Fans & Lights

Detailed Visualization

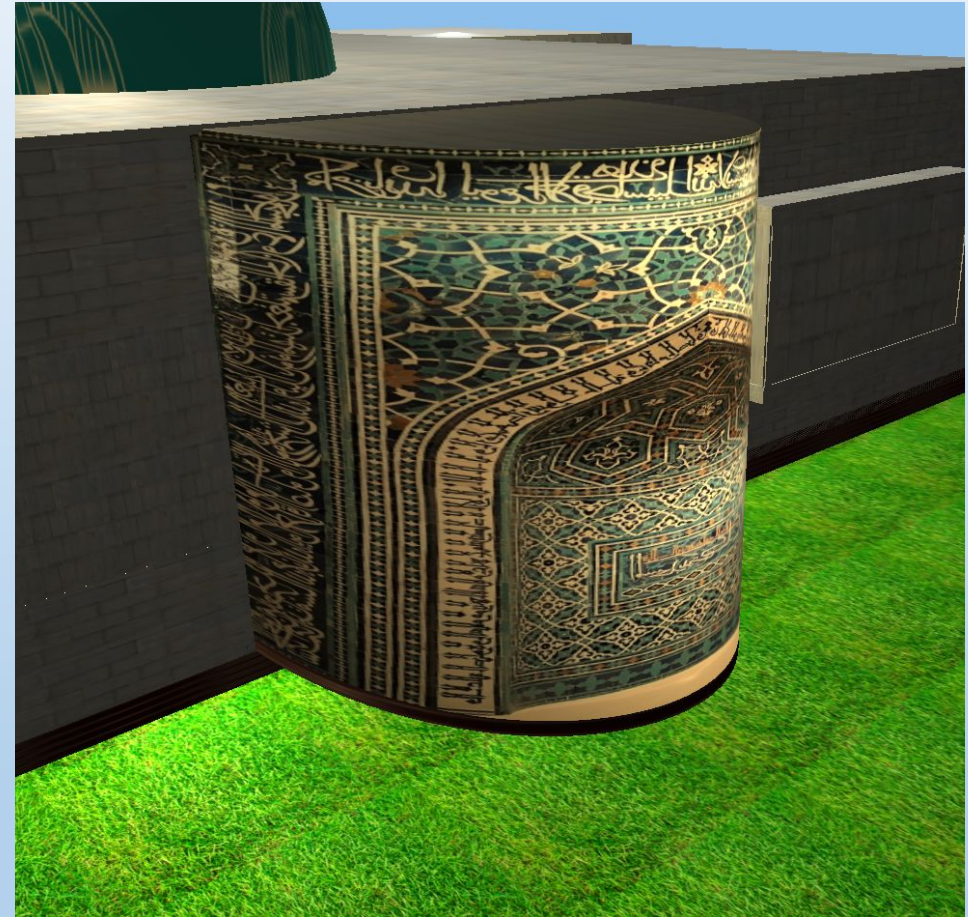


Fig-14: Mihrab

Detailed Visualization

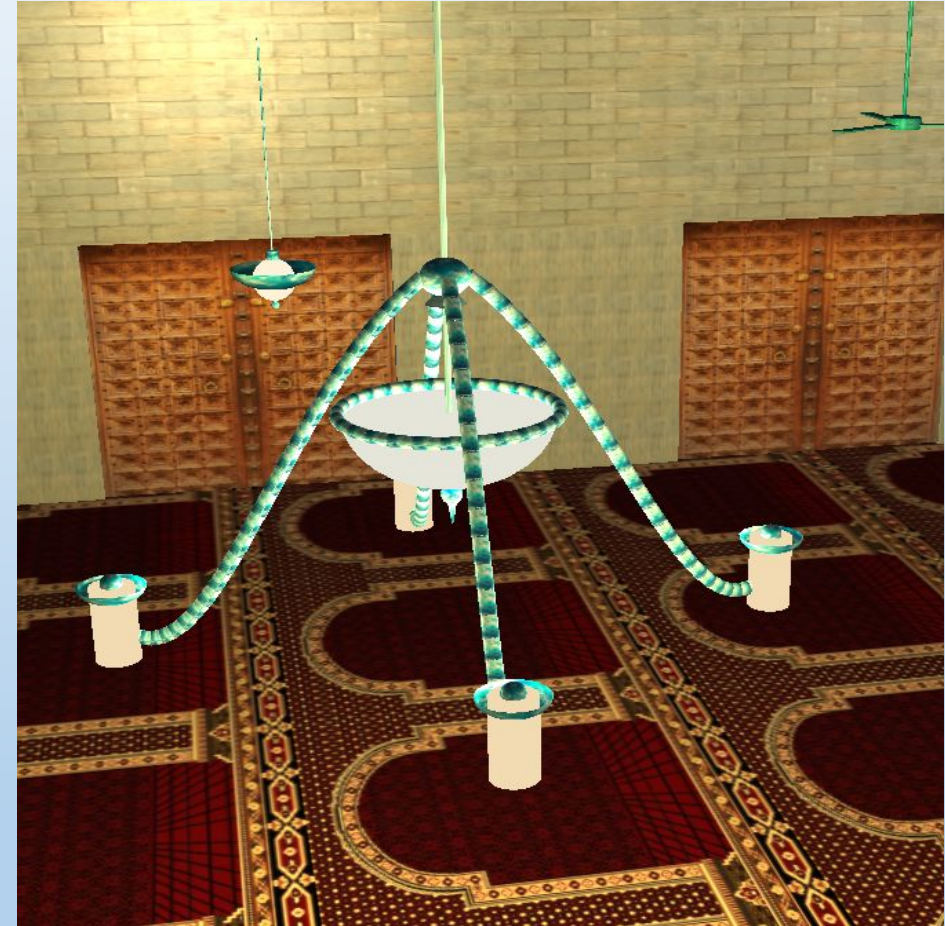
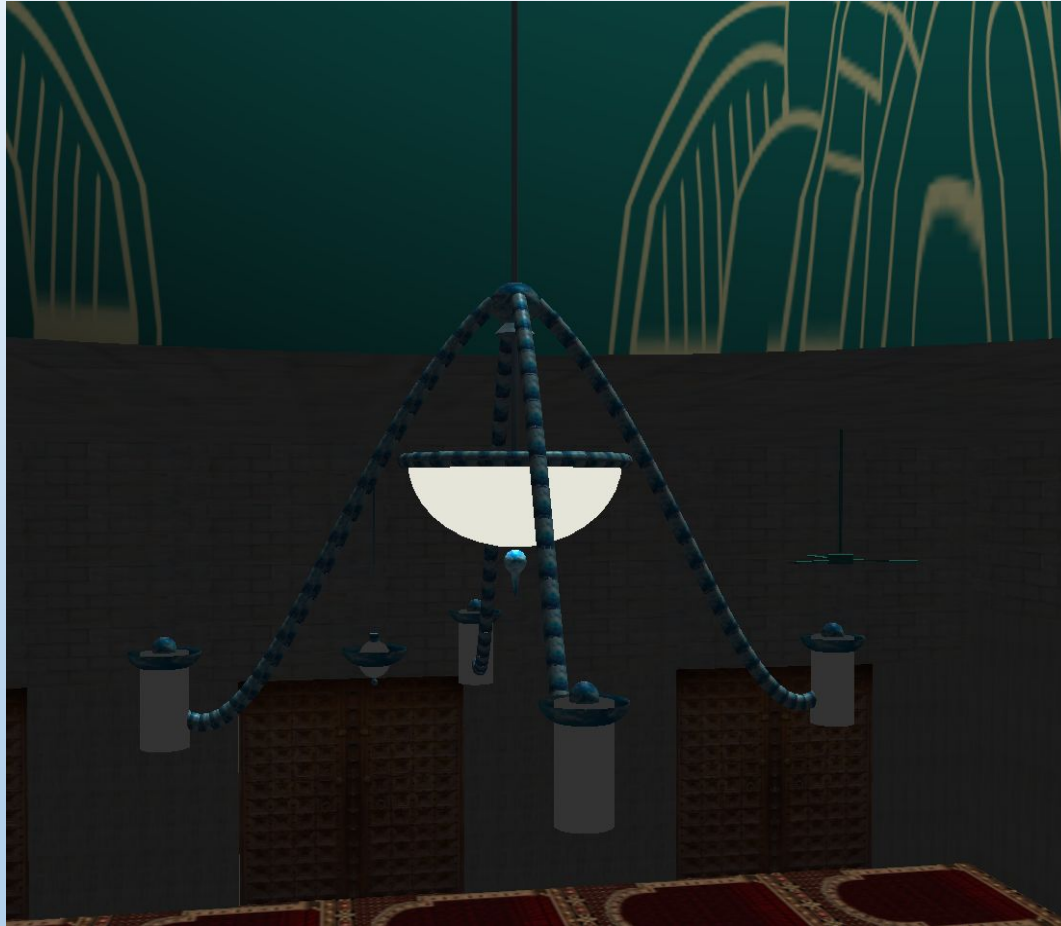


Fig-15: Chandelier

Detailed Visualization

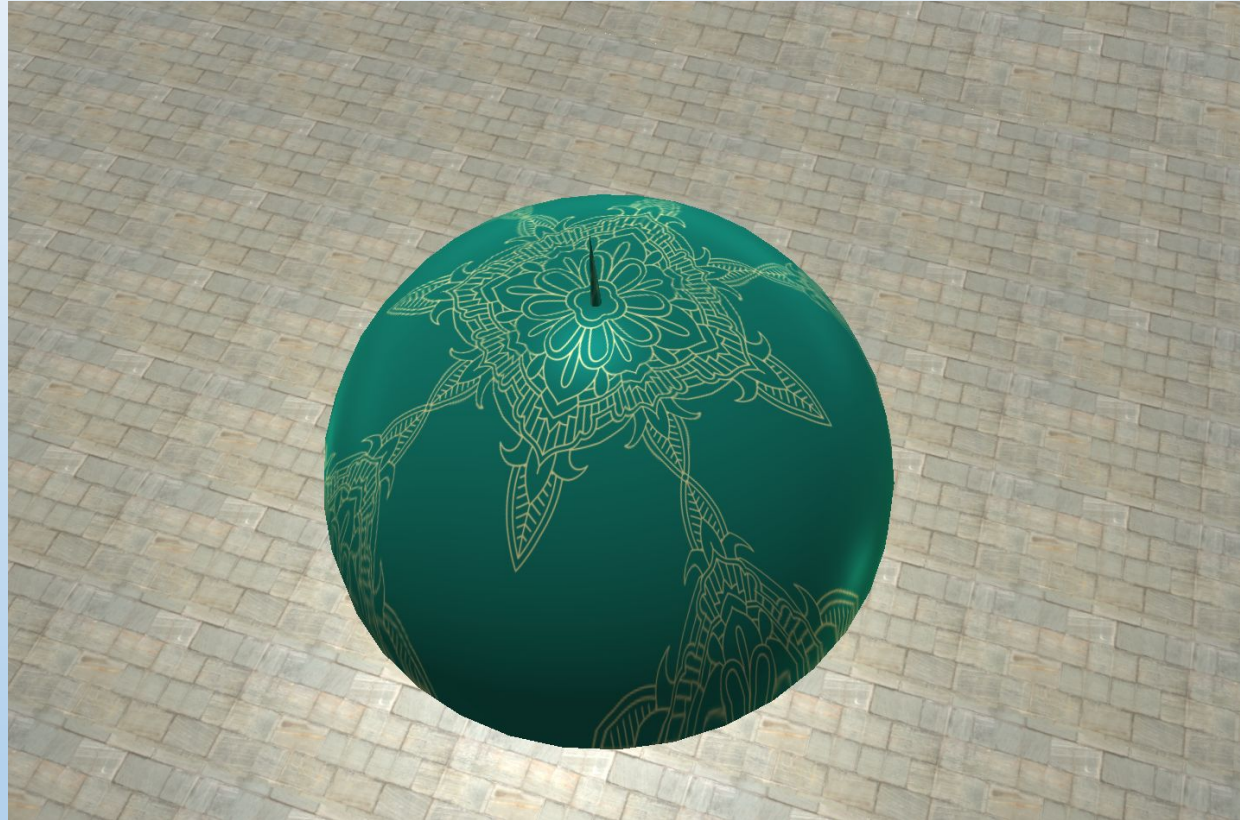


Fig-16: Dome

Detailed Visualization



Fig-17: Minaret

Conclusion

What I Have Implemented:

- Draw structure with Cube
- Draw structure with Bezier Curve
- Adding Texture Mapping on objects and surfaces
- Implementing Camera Movement and Rotation
- Implementing Roll, Pitch, Yaw camera control
- Implementing Spot Light, Directional Light, and Point Light
- Implementing Light On/Off interaction controls
- Creating and animating doors, windows, water droplets
- Creating Domes, minarets, chandelier, bookshelves, wahdu place
- Building scene environment elements (road, trees, pond, lamp posts)

ith Cube

Draw structure with Bezier Curve

Physics-based movement of water particles in the fountain

Adding Texture Mapping on objects and surfaces

Implementing Camera Movement and Rotation

**Thank
You**